1.18 Graph Sketching: Rational Functions

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GCSE RECAP: Reciprocal Graphs

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Sketch $y = \frac{1}{x}$

y

x

Sketch $y = -\frac{3}{x}$

y

x

Note: The scaling caused by the 3 isn't observable for this graph in isolation because the axes have no scale. This will only be observation for multiple graphs on the same axes.

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Reciprocal Graphs

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Sketch $y = \frac{3}{x^2}$

This is new to the A Level 2017 syllabus.

y

x

Sketch $y = -\frac{4}{x^2}$

y

x

Hint: Note that anything squared will always be at least 0.

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Reciprocal Graphs

23

On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$

y

$y = \frac{3}{x}$

The y value for $y = \frac{3}{x}$ will be 3 times greater than $y = \frac{1}{x}$

$y = \frac{1}{x}$ x

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1 Use a separate diagram to sketch each pair of graphs.

a $y = \frac{2}{x}$ and $y = \frac{4}{x}$

b $y = \frac{2}{x}$ and $y = -\frac{2}{x}$

c $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$

d $y = \frac{3}{x}$ and $y = \frac{8}{x}$

e $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$

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2 Use a separate diagram to sketch each pair of graphs. **a** $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$